

Hailey A. Reed

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Education

New York University, Courant Institute of Mathematical Sciences

New York, NY

Master of Science in Computer Science — Incoming Fall 2026 (Expected May 2028)

University of Connecticut

Storrs, CT

Bachelor of Science, Major in Computer Science, Minor in Mathematics • GPA: 3.65/4.0

2022-2026

• Concentration in Artificial Intelligence • University Honors Laureate • UConn STEM Scholar Community • Dean's List

Publications

Niu, Y., Reed, H. A., & Yang, Q. (2026). *DeltaSegment: Multifidelity Medical Image Segmentation via Change Detection*.

Proceedings of the 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI).

<https://doi.org/10.1109/ISBI61048.2026.11515780>

Research Experience

Undergraduate Research Assistant

Storrs, CT

Advisor: Dr. Qian Yang | Scientific & Computational Machine Learning Laboratory (SciMaLL)

Project: *DeltaSegment: Multifidelity Medical Image Segmentation via Change Detection*

Jan 2025 – May 2026

- Co-developed DeltaSegment, a multi-fidelity, change-detection-based Semi-Siamese U-Net framework that improves small-data medical image segmentation by leveraging high-low fidelity image pairs generated via color quantization.
- Presented research findings at the 2026 IEEE International Symposium on Biomedical Imaging (ISBI) in London, UK.

Honors Thesis: *Improving Medical Image Segmentation via Multi-Fidelity Modeling*

Aug 2025 – May 2026

- Designed and compared multiple U-Net-based architectures to identify effective formulations for exploiting multi-fidelity differencing as a boundary-aware learning signal in medical image segmentation.
- Investigated limitations of patch-based cross-validation evaluation, analyzing discrepancies between validation performance and held-out test generalization in biomedical imaging.

Undergraduate Research Assistant

Storrs, CT

Advisor: Professor Joseph Johnson

Project: *GDP Nowcasting using FRED-MD Macroeconomic Indicators*

May 2025 – Jan 2026

- Reproduced the GDP nowcasting framework from Németh & Hadházi (2023), implementing benchmark neural architectures (MLP, 1D-CNN, Elman RNN, LSTM, GRU) on the FRED-MD macroeconomic panel.
- Built a regime-switching ensemble that identifies macroeconomic conditions and selects the optimal model.

Leadership & Innovation

Entrepreneurial Lead – Accelerate UConn (NSF I-Corps) Cohort 32

Hartford, CT

UConn Center for Entrepreneurship & Innovation

June 2025 – July 2025

- Selected as Entrepreneurial Lead for the NSF I-Corps Northeast Regional cohort, translating lab-developed Semi-Siamese change-detection segmentation research into LitheVision, a deployable defect-detection platform for discrete-part manufacturing. Led customer discovery, stakeholder outreach, venture capital discussions, and commercialization analysis for AI-driven manufacturing inspection tools.

Projects

AlarmCast – Senior Design Project (Team Lead)

Sept 2025 – May 2026

- Led development of a Raspberry Pi smoke/CO alarm system with real-time mobile alerts using Flask, REST APIs, PostgreSQL/AWS infrastructure, Argon2-based authentication, and integrated hardware-mobile communication pipelines.

Personal Portfolio Website – <https://haileyannreed.github.io/portfolio/>

Mar 2025 – Present

Technical Skills

Languages: Python, C, SQL, Haskell, JavaScript, HTML/CSS

ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Ray Tune, ASHA

ML Engineering & Training: Computer Vision (Image Segmentation, Change Detection, Multi-Fidelity Modeling, Zero-/Few-Shot Learning, Data Augmentation), Time-Series Forecasting, Model Evaluation

Databases & Infrastructure: PostgreSQL, AWS (EC2, Aurora), Flask, REST APIs, Git, HPC, LaTeX